IGCSE 0607 Practice Questions — E5.1 — Combined Questions & Markscheme

Geometrical terms

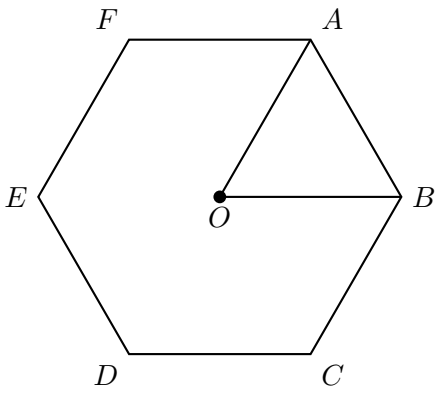
Total Questions: 24 — **Total Marks:** 144 Marks

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- **E5.1** Geometrical terms
  → **Paper 2 (Non-Calculator)** *(15 Qs)* · **Paper 4 (Calculator)** *(9 Qs)*

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1 The diagram shows a regular polygon $ABCDEF$ with centre O.



(a) Write down the mathematical name of the polygon $ABCDEF$.

..... **[1]**

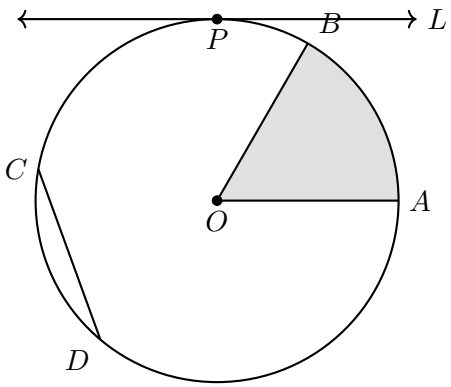
(b) Write down the mathematical name for the angle AFE.

..... **[1]**

(c) Write down the mathematical name for the triangle AOB.

..... **[1]**

2 The diagram shows a circle with centre O.
The straight line L touches the circle at the point P.



(a) Write down the mathematical name for the straight line CD.

..... **[1]**

(b) Write down the mathematical name for the straight line L.

..... **[1]**

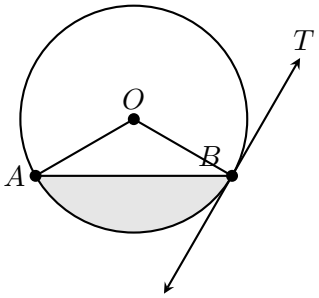
(c) Write down the mathematical name for the shaded region OAB.

..... **[1]**

(d) The angle AOB is $60°$.
Write down the mathematical name for this type of angle.

..... **[1]**

3 The diagram shows a circle, centre O.
A and B are points on the circumference of the circle.
The line passing through T touches the circle at B.



(a) Write down the geometrical name of the line AB.

..... **[1]**

(b) Write down the geometrical name of the line BT.

..... **[1]**

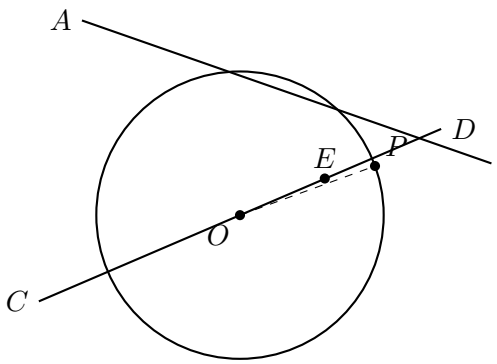
(c) Write down the geometrical name of the shaded region.

..... **[1]**

(d) Write down the mathematical name of triangle OAB.

..... **[1]**

4 The diagram shows a circle with centre O.
The lines AB and CD intersect at E.
AB touches the circle at P, and CD passes through the centre O.



(a) Write down the mathematical name for the line segment OP.

..... **[1]**

(b) Write down the mathematical name for the line AB.

..... **[1]**

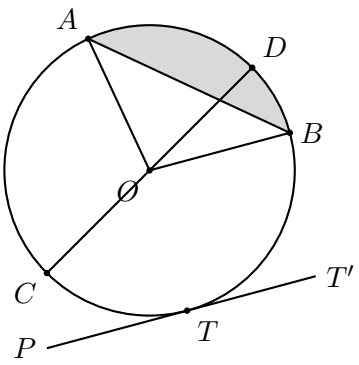
(c) Write down the mathematical name for the line segment CD.

..... **[1]**

(d) State the size of angle OPT.

..... ° **[1]**

5 The diagram shows a circle with centre O and points A, B, C and D on its circumference.
The line PT is a tangent to the circle at T, and the shaded region is bounded by the chord AB and the minor arc AB.



(a) Write down the geometrical name for:

(i) the line segment CD,

..... **[1]**

(ii) the shaded region bounded by chord AB and minor arc AB,

..... **[1]**

(iii) the region bounded by radii OA, OB and the major arc AB.

..... **[1]**

(b) State the type of angle formed by the reflex angle $\angle AOB$ if the acute angle $\angle AOB = 100°$.

..... **[1]**

6 **(a)** **(i)** A polygon has interior angles that sum to 1080°.

Work out the number of sides of this polygon.

..... **[2]**

(ii) Write down the mathematical name of this polygon.

..... **[1]**

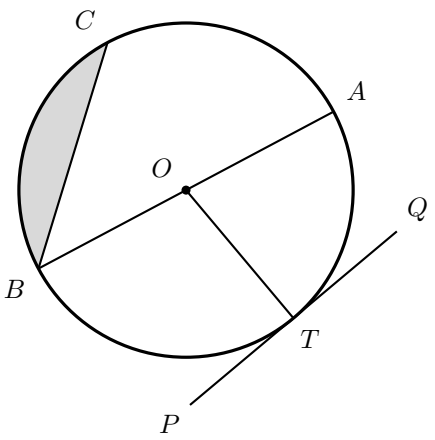
(b) Write down the mathematical name of a quadrilateral with all four sides equal in length and no right angles.

..... **[1]**

(c) Write down the mathematical name of a 3D solid that has a uniform cross-section and two parallel circular faces.

..... **[1]**

7 The diagram shows a circle, centre O, with diameter AB.
The points A, B, C, and T lie on the circumference of the circle.
The straight line PQ touches the circle at T.



(a) Write down the mathematical name for

(i) the straight line PQ,

..... **[1]**

(ii) the straight line BC,

..... **[1]**

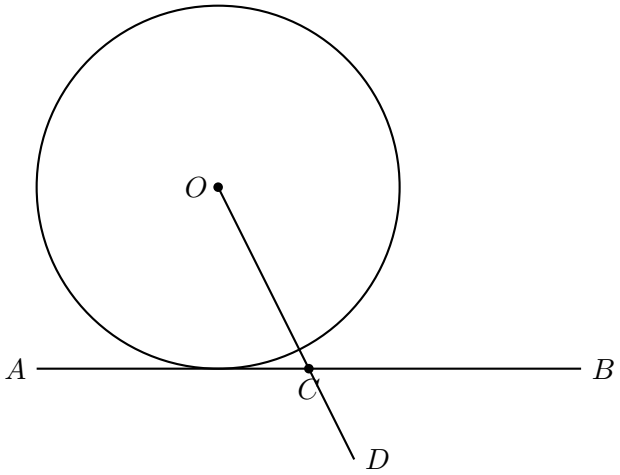
(iii) the shaded region bounded by BC and the minor arc BC.

..... **[1]**

(b) State the geometrical word that describes the relationship between radius OT and line PQ, and write down the size of angle OTQ.

Relationship: ..... , Angle $OTQ =$ ..... **[2]**

8 The diagram shows a circle with centre O.
The line AB touches the circle at the point C.
The line OD passes through C.



(a) Write down the mathematical name for the straight line AB.

..... **[1]**

(b) Write down the mathematical name for the straight line OC.

..... **[1]**

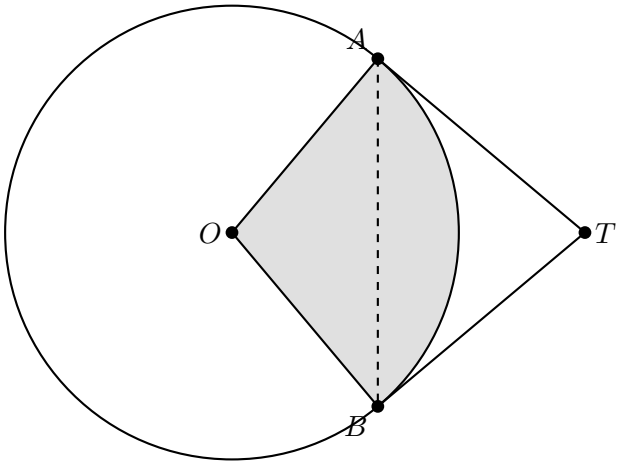
(c) Write down the size of angle OCB.

Angle OCB = ... ° **[1]**

(d) The length of OC is 5 cm.
Calculate the circumference of the circle.
Give your answer in terms of π.

..... cm **[2]**

9 Points A and B lie on a circle, centre O.
The lines TA and TB are tangents to the circle.
The region enclosed by the arc AB and the radii OA and OB is shaded.



(a) Write down the mathematical name for the quadrilateral $OATB$.

..... **[1]**

(b) Write down the mathematical name for the shaded region.

..... **[1]**

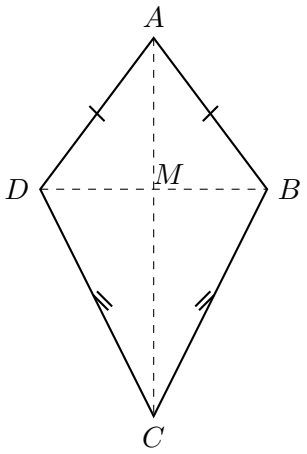
(c) Write down the mathematical name for the straight line AB.

..... **[1]**

(d) Triangle OAT and triangle OBT are exactly the same size and shape.
Write down the mathematical term for two identical shapes.

..... **[1]**

10 The diagram shows a quadrilateral $ABCD$.
$AB = AD$ and $CB = CD$.
The diagonals AC and BD intersect at M.



(a) Write down the mathematical name of the quadrilateral $ABCD$.

..... **[1]**

(b) Write down the mathematical term that describes the relationship between the line AC and the line BD.

..... **[1]**

(c) Triangle ABM and Triangle ADM are identical in shape and size.
Write down the mathematical word for this.

..... **[1]**

(d) The angle ABC is $110°$.
Write down the mathematical name for this type of angle.

..... **[1]**

(e) Write down the mathematical name for the line segment AC in relation to the line segment BD.

..... **[1]**

11 A regular polygon has n sides.
Each interior angle of this polygon is 156°.

(a) Work out the size of one exterior angle of this polygon.

..... ° **[1]**

(b) Find the value of n.

$n =$ ..... **[2]**

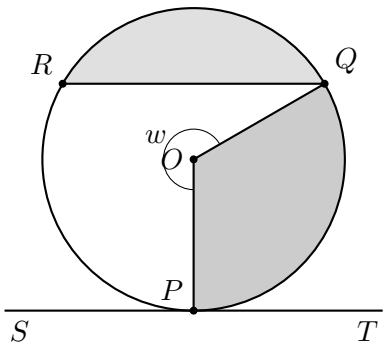
(c) Write down the mathematical name for a regular polygon with 10 sides.

..... **[1]**

(d) State whether a regular 15-sided polygon has an acute, obtuse, or reflex interior angle.

..... **[1]**

12 The diagram shows a circle, centre O.
P, Q and R are points on the circumference.
The straight line ST touches the circle at P.



(a) Write down the mathematical name of the line ST.

..... **[1]**

(b) Write down the mathematical name of the line segment QR.

..... **[1]**

(c) Write down the mathematical name of the shaded region bounded by the line QR and the minor arc QR.

..... **[1]**

(d) Write down the mathematical name of the shaded region bounded by the radii OP and OQ and the minor arc PQ.

..... **[1]**

(e) The angle marked w is greater than $180°$ and less than $360°$.
Write down the geometrical term for this type of angle.

..... **[1]**

13 Quadrilateral $PQRS$ has the following geometrical properties:

- Only one pair of opposite sides are parallel.
- The non-parallel sides are equal in length.
- It has one line of symmetry and no rotational symmetry.

(a) Write down the full geometrical name of quadrilateral $PQRS$.

..... **[1]**

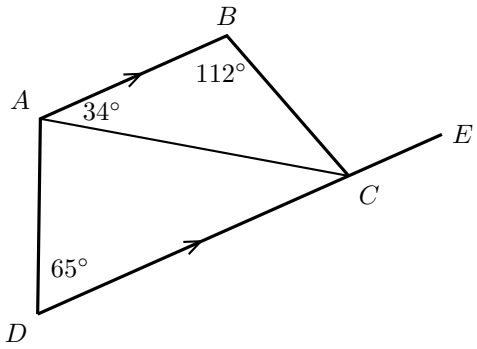
(b) A regular polygon has an exterior angle of $36°$.
State the geometrical name of this regular polygon.

..... **[2]**

(c) A solid prism has a regular decagon as its cross-section.
Write down the number of faces and the number of edges of this prism.

Faces = ... , Edges = ... **[2]**

14 In the diagram, $ABCD$ is a quadrilateral with AB parallel to DC.
The side DC is extended to E, and AC is a diagonal.
Angle $ADC = 65°$, angle $BAC = 34°$, and angle $ABC = 112°$.



(a) Write down the special mathematical name for quadrilateral $ABCD$.

..... **[1]**

(b) For vertex B,

(i) write down the geometrical term that describes the interior angle ABC,

..... **[1]**

(ii) calculate the size of the reflex angle at vertex B.

..... ° **[1]**

(c) Calculate the size of

(i) angle ACD,

Angle $ACD =$ ..... ° **[1]**

(ii) the exterior angle BCE.

Angle $BCE =$ ..... ° **[2]**

15 **(a)** The interior angle of a regular polygon is 144°.

(i) Work out the number of sides of this polygon.

..... **[2]**

(ii) Write down the geometrical name of this polygon.

..... **[1]**

(b) A hexagon has interior angles of $x°$, $2x°$, $(x + 30)°$, $(3x + 60)°$, 110° and 100°.

(i) Calculate the value of x.

$x =$ ..... **[3]**

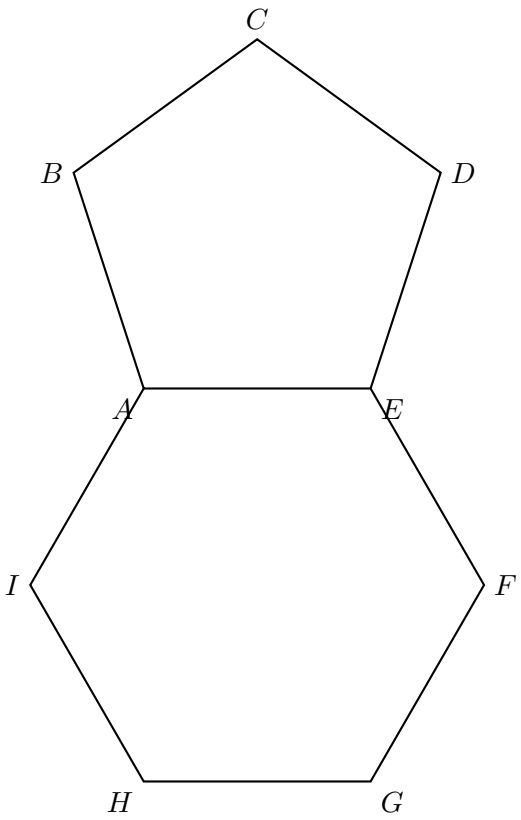
(ii) Write down the geometrical name for an angle of $(x + 30)°$.

..... **[1]**

(iii) Write down the geometrical name for an angle of $(3x + 60)°$.

..... **[1]**

16 The diagram shows a regular pentagon $ABCDE$ and a regular hexagon $AEFGHI$ joined along the edge AE.



(a) Calculate the size of an interior angle of the regular pentagon $ABCDE$.

..... ° **[2]**

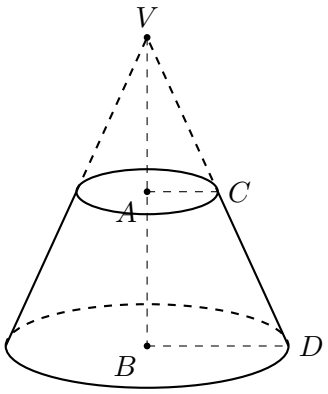
(b) Calculate the size of an interior angle of the regular hexagon $AEFGHI$.

..... ° **[2]**

(c) Calculate the size of the reflex angle BAI.

..... ° **[3]**

17 The diagram shows a solid frustum formed by removing a smaller right circular cone from the top of a larger right circular cone.
The plane of the circular top face is parallel to the plane of the circular base.



(a) State the geometrical relationship between the original cone with vertex V and the removed cone with vertex V.

..... **[1]**

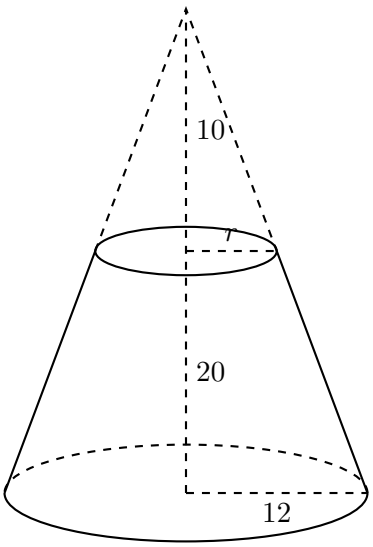
(b) Explain why $\triangle VAC$ is a right-angled triangle and why AC is parallel to BD.

..... **[2]**

(c) Given that $AC = 4.5$ cm, $BD = 7.5$ cm and the perpendicular height of the frustum $AB = 6$ cm, calculate the perpendicular height VA of the removed cone.

$VA =$ ..... cm **[3]**

18 The diagram shows a large solid cone with a smaller solid cone removed from the top.
The cut is made parallel to the base of the large cone.
All dimensions are in centimetres.



(a) Write down the mathematical name of the solid with height 20.

..... **[1]**

(b) The small cone is geometrically related to the original large cone.

(i) Write down the mathematical word that describes the relationship between the small cone and the large cone.

..... **[1]**

(ii) Work out the value of the radius, r.

$r =$ ..... **[2]**

(iii) Calculate the slant edge of the solid with height 20.

..... *cm* **[3]**

19 A map shows three towns A, B and C.
Town B is 45 km from A on a bearing of 062°.
Town C is 45 km from A on a bearing of 138°.

(a) Write down the mathematical name of triangle ABC.

..... **[1]**

(b) Work out the bearing of A from C.

..... ° **[2]**

(c) **(i)** Point M is the midpoint of the straight line BC.
Write down the mathematical name of the line AM in relation to the line BC.

..... **[1]**

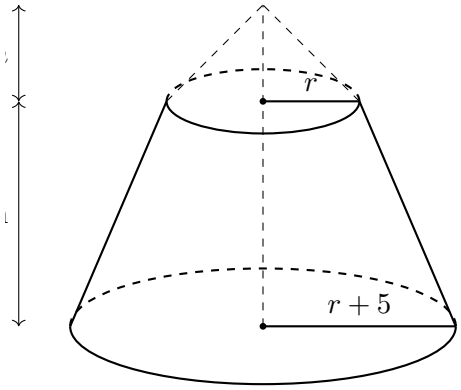
(ii) Calculate the bearing of M from A.

..... ° **[2]**

(d) A fourth town D is located such that the line AB is parallel to the line DC and the line AD is not parallel to the line BC.
Write down the mathematical name of the quadrilateral $ABCD$.

..... **[1]**

20 The diagram shows a solid frustum obtained by cutting a smaller right circular cone of height h cm from a larger right circular cone of height $(h + 12)$ cm.
The radius of the top circular face is r cm and the radius of the bottom circular base is $(r + 5)$ cm.
The two cones are mathematically similar.



(a) Given that the scale factor relating the linear dimensions of the smaller cone to the larger cone is $\frac{3}{5}$, show that $h = 18$.

[3]

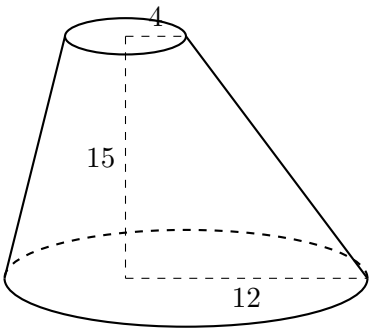
(b) Find the value of r.

$r =$ ..... **[2]**

(c) The ratio of the volume of the smaller cone to the volume of the frustum is $27 : k$.
Find the value of k.

$k =$ ..... **[2]**

21 The diagram shows a frustum of a cone.
The radius of the circular base is 12 cm and the radius of the circular top is 4 cm.
The vertical height of the frustum is 15 cm.



(a) Calculate the volume of the frustum.
The volume, V, of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.

..... cm^3 **[5]**

(b) A different shape is geometrically similar to the frustum.
The ratio of the surface area of this shape to the surface area of the frustum is 9 : 25.
Calculate the vertical height of this shape.

..... cm **[3]**

22 A regular polygon has n sides.

(a) The interior angle of this regular polygon is 4 times the size of its exterior angle.
Calculate the value of n and write down the mathematical name of this polygon.

$n =$ ..... , Name = ... **[4]**

(b) A solid uniform prism has a cross-section in the shape of this regular polygon.
Write down the number of faces, edges, and vertices of this prism.

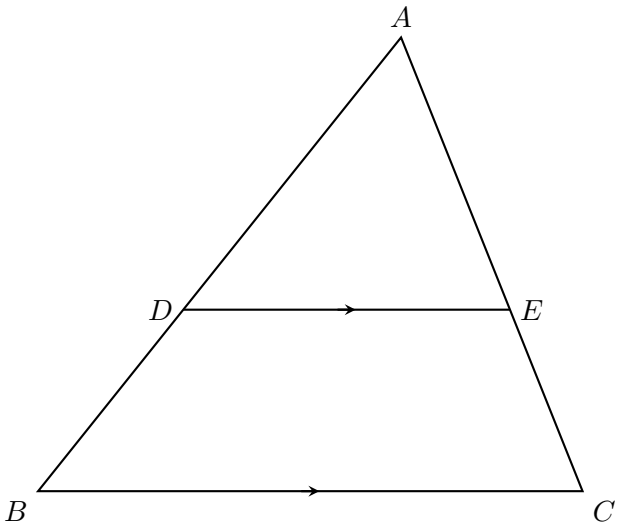
Faces = ... , Edges = ... , Vertices = ... **[3]**

(c) Two mathematically similar cones have base areas in the ratio 9 : 16.
The volume of the smaller cone is 108π cm^3.

Calculate the volume of the larger cone.
Give your answer in terms of π.

..... cm^3 **[3]**

23 The diagram shows triangle ABC.
D is a point on AB and E is a point on AC such that DE is parallel to BC.
$AD = 6$ cm, $DB = 4$ cm, $AE = 7.5$ cm and $BC = 15$ cm.



(a) Explain why triangle ADE is similar to triangle ABC.

[2]

(b) Calculate the length of DE.

..... cm **[2]**

(c) Calculate the length of EC.

..... cm **[2]**

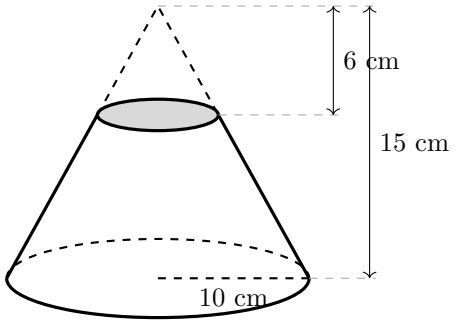
(d) State the geometrical name of the quadrilateral $DBCE$.

..... **[1]**

(e) The area of triangle ADE is 18 cm^2.
Calculate the area of the quadrilateral $DBCE$.

..... cm^2 **[3]**

24 The diagram shows a solid formed by cutting a smaller right cone of height 6 cm from the top of a larger right cone of height 15 cm and base radius 10 cm. The cut is made along a plane parallel to the circular base.



(a) Write down the mathematical name for

(i) the remaining solid after the top cone is removed,

..... **[1]**

(ii) a solid pyramid that has 4 triangular faces, 6 edges, and 4 vertices.

..... **[1]**

(b) Explain why the removed cone is similar to the original large cone, and write down the scale factor that maps the original large cone onto the removed cone.

Reason: ..... , Scale factor = ... **[2]**

(c) Calculate the radius of the upper circular face of the remaining solid.

..... cm **[2]**

(d) A regular polygon has an exterior angle of 36°.

(i) Write down the mathematical name of this regular polygon.

..... **[2]**

(ii) A right prism has this regular polygon as its cross-section.
State the total number of faces and the total number of edges of this prism.

Faces = ... , Edges = ... **[2]**

1 **(a)** Hexagon **[B1]**

(b) Obtuse **[B1]**

(c) Equilateral **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** Chord **[B1]**

(b) Tangent **[B1]**

(c) Sector **[B1]**

(d) Acute **[B1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** Chord **[B1]**

(b) Tangent **[B1]**

(c) Minor segment (Accept segment) **[B1]**

(d) Isosceles **[B1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Radius **[B1]**

(b) Tangent **[B1]**

(c) Diameter **[B1]**

(d) 90 **[B1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** **(i)** Diameter **[B1]**

(ii) Minor segment (or segment) **[B1]**

(iii) Major sector **[B1]**

(b) Reflex angle (or 260°, reflex) **[B1]**

[Back to Question] *[Total Marks: 4]*

6 **(a)** **(i)** $(n-2) \times 180 = 1080$ or $1080 \div 180$ **[M1]**

8 **[A1]**

(ii) Octagon **[B1]**

(b) Rhombus **[B1]**

(c) Cylinder **[B1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** **(i)** tangent **[B1]**

(ii) chord **[B1]**

(iii) minor segment (accept segment) **[B1]**

(b) Relationship: perpendicular **[B1]**

Angle $OTQ = 90°$ **[B1]**

[Back to Question] *[Total Marks: 5]*

8 **(a)** Tangent **[B1]**

(b) Radius **[B1]**

(c) 90 **[B1]**

(d) Uses $2 \times \pi \times 5$ **[M1]**

10π **[A1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** Kite **[B1]**

(b) Sector (or minor sector) **[B1]**

(c) Chord **[B1]**

(d) Congruent **[B1]**

[Back to Question] *[Total Marks: 4]*

10 **(a)** Kite **[B1]**

(b) Perpendicular **[B1]**

(c) Congruent **[B1]**

(d) Obtuse **[B1]**

(e) Perpendicular bisector **[B1]**

[Back to Question] *[Total Marks: 5]*

11 **(a)** $180 - 156 = 24$ **[B1]**

(b) $\frac{360}{24}$ **[M1]**

15 **[A1]**

(c) Decagon **[B1]**

(d) Obtuse **[B1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** Tangent **[B1]**

(b) Chord **[B1]**

(c) Segment **[B1]**

(d) Sector **[B1]**

(e) Reflex **[B1]**

[Back to Question] *[Total Marks: 5]*

13 **(a)** Isosceles trapezium **[B1]**

(b) Calculates number of sides: $\frac{360^\circ}{36^\circ} = 10$ **[M1]**

Regular decagon (or decagon) **[A1]**

(c) Faces = 12 **[B1]**

Edges = 30 **[B1]**

[Back to Question] *[Total Marks: 5]*

14 **(a)** trapezium **[B1]**

(b) **(i)** obtuse **[B1]**

(ii) $360 - 112 = 248$ **[B1]**

(c) **(i)** 34 **[B1]**

(ii) Angle $BCA = 180^\circ - 112^\circ - 34^\circ = 34^\circ$ (or interior angle $BCD = 68^\circ$) **[M1]**

112 **[A1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** **(i)** Exterior angle $= 180^\circ - 144^\circ = 36^\circ$ **[M1]**

$\frac{360}{36} = 10$ **[A1]**

(ii) Decagon **[B1]**

(b) **(i)** Sum of interior angles $= (6-2) \times 180^\circ = 720^\circ$ **[B1]**

$x + 2x + (x + 30) + (3x + 60) + 110 + 100 = 720$ OR $7x + 300 = 720$ **[M1]**

$x = 60$ **[A1]**

(ii) Right angle (since $60 + 30 = 90$) **[B1]**

(iii) Reflex angle (since $3(60) + 60 = 240$) **[B1]**

[Back to Question] *[Total Marks: 8]*

16 **(a)** $\frac{(5-2)\times 180}{5}$ or $180 - \frac{360}{5}$ **[M1]**

108 **[A1]**

(b) $\frac{(6-2)\times 180}{6}$ or $180 - \frac{360}{6}$ **[M1]**

120 **[A1]**

(c) Identifies angles around point A: $108°$ and $120°$ **[M1]**

Calculates reflex angle $= 360 - 108 - 120$ **[M1]**

132 **[A1]**

[Back to Question] *[Total Marks: 7]*

17 **(a)** Similar (geometrically similar) **[B1]**

(b) In a right circular cone, the axis VB is perpendicular to the circular base planes, so $\angle VAC = 90°$ **[B1]**

Both AC and BD lie in parallel planes and are perpendicular to the same line VB in the same cross-sectional plane, so AC is parallel to BD **[B1]**

(c) Uses similarity scale factor: $\frac{VA}{VA+6} = \frac{4.5}{7.5}$ (or $\frac{3}{5}$) **[M1]**

$5VA = 3(VA + 6)$

$2\text{VA} = 18[M1]$

9 **[A1]**

[Back to Question] *[Total Marks: 6]*

18 **(a)** Frustum **[B1]**

(b) **(i)** Similar **[B1]**

(ii) Scale factor $= \frac{20+10}{10} = 3$ or $\frac{10}{30} = \frac{1}{3}$ **[M1]**

$r = \frac{12}{3} = 4$ **[A1]**

(iii) Radius difference $= 12 - 4 = 8$ **[M1]**

$\sqrt{20^2 + 8^2}$ **[M1]**

21.5 or 21.54... **[A1]**

[Back to Question] *[Total Marks: 7]*

19 **(a)** Isosceles **[B1]**

(b) $138 + 180$ or $360 - (180 - 138)$ **[M1]**

318 **[A1]**

(c) **(i)** Perpendicular bisector **[B1]**

(ii) Angle $BAC = 138 - 62 = 76°$ **[M1]**

Bearing $= 062 + (76 \div 2) = 100°$ **[A1]**

(d) Trapezium **[B1]**

[Back to Question] *[Total Marks: 7]*

20 **(a)** $\dfrac{h}{h+12} = \dfrac{3}{5}$ **[M1]**

$5h = 3(h + 12)$ **[M1]**

$5h = 3h + 36 \implies 2h = 36 \implies h = 18$ **[A1]**

(b) $\dfrac{r}{r+5} = \dfrac{3}{5}$ or $\dfrac{r}{r+5} = \dfrac{18}{30}$ **[M1]**

$5r = 3r + 15 \implies 2r = 15 \implies r = 7.5$ **[A1]**

(c) Volume ratio (small cone : large cone) $= \left(\dfrac{3}{5}\right)^3 = \dfrac{27}{125}$ **[M1]**

Volume of frustum $= 125 - 27 = 98$, so $k = 98$ **[A1]**

[Back to Question] *[Total Marks: 7]*

21 **(a)** Uses similar triangles to find the height H of the full cone: $\dfrac{H}{12} = \dfrac{H-15}{4}$ **[M1]**

$4H = 12H - 180 \implies 8H = 180 \implies H = 22.5$ cm **[A1]**

Volume of full cone $= \dfrac{1}{3} \times \pi \times 12^2 \times 22.5 = 1080\pi$ **[M1]**

Volume of small cone removed $= \dfrac{1}{3} \times \pi \times 4^2 \times (22.5 - 15) = \dfrac{1}{3} \times \pi \times 16 \times 7.5 = 40\pi$ **[M1]**

Volume of frustum $= 1080\pi - 40\pi = 1040\pi$ or 3267 or 3270 (3 sf) **[A1]**

(b) Identifies the area scale factor is $\dfrac{9}{25}$ **[B1]**

Calculates the length scale factor: $\sqrt{\dfrac{9}{25}} = \dfrac{3}{5}$ **[M1]**

New height $= 15 \times \dfrac{3}{5} = 9$ **[A1]**

[Back to Question] *[Total Marks: 8]*

22 **(a)** Setting up an equation connecting interior and exterior angles, e.g., $4x + x = 180$ **[M1]**

Exterior angle $= 36°$ or Interior angle $= 144°$ **[A1]**

$n = 10$ **[A1]**

Decagon **[B1]**

(b) Faces = 12 **[B1]**

Edges = 30 **[B1]**

Vertices = 20 **[B1]**

(c) Finding the linear scale factor $\sqrt{\frac{16}{9}}$ or $\frac{4}{3}$ **[M1]**

Finding the volume scale factor $\left(\frac{4}{3}\right)^3$ or $\frac{64}{27}$ **[M1]**

256π **[A1]**

[Back to Question] *[Total Marks: 10]*

23 **(a)** Angle ADE = Angle ABC because they are corresponding angles **[B1]**

Angle A is common to both triangles OR Angle AED = Angle ACB (corresponding angles) **[B1]**

(b) Length scale factor $= \frac{6}{10}$ or 0.6 **[M1]**

$DE = 15 \times 0.6 = 9$ **[A1]**

(c) $AC = \frac{7.5}{0.6} = 12.5$ **[M1]**

$EC = 12.5 - 7.5 = 5$ **[A1]**

(d) Trapezium **[B1]**

(e) Area scale factor $= 0.6^2 = 0.36$ **[M1]**

Area of triangle $ABC = \frac{18}{0.36} = 50$ **[M1]**

Area of $DBCE = 50 - 18 = 32$ **[A1]**

[Back to Question] *[Total Marks: 10]*

24 **(a)** **(i)** frustum **[B1]**

(ii) tetrahedron **[B1]**

(b) Reason: the cut is parallel to the base so corresponding angles are equal **[B1]**

Scale factor = 0.4 (or $\frac{2}{5}$) **[B1]**

(c) $10 \times \frac{6}{15}$ **[M1]**

4 **[A1]**

(d) **(i)** Number of sides $= \frac{360}{36} = 10$ **[M1]**

decagon **[A1]**

(ii) Faces = 12 **[B1]**

Edges = 30 **[B1]**

[Back to Question] *[Total Marks: 10]*